Notes: Hansen & Richard (Econometrica, 1987)

The Role of Conditioning Information in Deducing Testable Restrictions Implied by Dynamic Asset Pricing Models

Contents

1 Big picture

- **Question.** How does conditioning information affect the testable restrictions implied by dynamic asset pricing models?
- **Central point.** In dynamic asset pricing models, both payoffs and prices should be treated as random variables. Payoffs are measurable with respect to future information, while prices are measurable with respect to current information.
- **Pricing-function view.** An asset pricing model is summarized by a pricing function π mapping a payoff p into a price $\pi(p)$ that is known at the trading date.
- **Mathematical contribution.** The paper develops a conditional Hilbert space representation of pricing functions. Under value additivity and continuity, each pricing function has a conditional inner-product representation.
- **Key representation.** There is a unique benchmark payoff p^* such that

$$\pi(p) = E(pp^* \mid G) = (p \mid p^*)_G.$$

- **Economic interpretation.** When no-arbitrage holds, p^* can be interpreted as a conditional stochastic discount factor or intertemporal marginal rate of substitution.
- **Mean-variance result.** Omitting conditioning information changes the mean-variance frontier. Returns on the unconditional frontier are also conditionally frontier returns, but many conditional frontier returns are not on the unconditional frontier.
- **Risk-free implication.** A conditionally risk-free return need not be on the unconditional mean-variance frontier unless the risk-free rate is constant.
- **Beta representation.** Testing an unconditional single-beta representation tests whether a return is on the unconditional frontier, not whether it is on the conditional frontier.
- **Econometric contribution.** The paper constructs a pseudo-pricing function $\pi^*(p) = E[\pi(p)]$ that maps payoffs into real numbers and gives restrictions expressible in unconditional moments.
- **Connection to empirical work.** The pseudo-pricing representation formalizes why Hansen–Singleton style tests use unconditional moment restrictions with instruments that are functions of conditioning information.

2 Data generation and measurement

2.1 Stationary and ergodic time series

- The paper begins with a probability space $(\Omega, \mathcal{F}, Pr)$ and a measure-preserving transformation S.
- If ω is the initial state, then $S^t(\omega)$ is the state at date t.
- A vector of observations is generated by

$$x_t(\omega) = x[S^t(\omega)].$$

- If S is ergodic, time-series averages converge to population expectations.

Interpretation: This setup gives a formal link between theoretical random variables and empirical time-series averages. The econometrician estimates unconditional moments even though agents condition on richer information.

2.2 Information sets

- Let G be information available at date zero.
- The date-t information set is

$$G_t = \{A_t : A_t = S^{-t}(A) \text{ for some } A \in G\}.$$

- The information sets are nondecreasing: agents learn over time.
- Prices at date t are measurable with respect to G_t.

Interpretation: A price is not deterministic from the econometrician's perspective. It is random, but it is constrained to be observable to traders at the time of trading.

2.3 Payoffs and prices

A one-period payoff sequence is generated as

$$p_{t+1}(\omega) = p[S^t(\omega)]. \tag{1}$$

The corresponding price sequence is

$$\pi_t(p_{t+1})(\omega) = \pi(p)[S^t(\omega)]. \tag{2}$$

Interpretation: The analysis can focus on date zero because stationarity translates the same pricing logic across time. This is why restrictions on unconditional moments can be connected to time-series averages.

3 Conditional Hilbert-space pricing

3.1 Payoff space and conditional inner product

The paper defines the conditional square-integrable payoff space as

$$P^+ = \{p \in I_1 : E(p^2 \mid G) < \infty\}. \tag{3}$$

For $p_1, p_2 \in P^+$, define

$$(p_1 \mid p_2)_G = E(p_1 p_2 \mid G), \tag{4}$$

and the conditional norm

$$\|p\|_G = [(p \mid p)_G]^{1/2}. \tag{5}$$

Interpretation: This is the conditional analogue of a Hilbert space. Inner products and norms are themselves random variables measurable with respect to the trading information G.

3.2 Pricing function assumptions

The pricing function $\pi : P \to I$ satisfies:

- **conditional linearity/value additivity:**

  $$\pi(w_1 p_1 + w_2 p_2) = w_1 \pi(p_1) + w_2 \pi(p_2)$$

  for $w_1, w_2 \in I$;
- **conditional continuity:** conditionally small payoffs have small prices;
- **nontriviality:** some payoff has nonzero price with probability one.

Interpretation: Value additivity says prices are linear in portfolios. Conditional linearity is stronger than ordinary linearity because portfolio weights can depend on current information.

3.3 Conditional Riesz representation

The main representation result is:

$$\pi(p) = (p \mid p^*)_G \quad \text{for all } p \in P, \tag{6}$$

for a unique payoff $p^* \in P$.

Interpretation: The entire asset pricing model can be represented by one benchmark payoff p^*. This is the conditional version of the stochastic discount factor representation.

3.4 No-arbitrage interpretation

When $P = P^+$ and the pricing function has no arbitrage opportunities,

$$Pr\{p^* > 0\} = 1. \tag{7}$$

Interpretation: A positive p^* acts like an equilibrium intertemporal marginal rate of substitution. It prices all payoffs by conditional covariance with payoffs.

4 Mean-variance implications of omitting conditioning information

4.1 Returns and zero-price payoffs

Define

$$R = \{p \in P : \pi(p) = 1\}, \tag{8}$$
$$Z = \{p \in P : \pi(p) = 0\}. \tag{9}$$

R is the set of all one-period returns; Z is the set of zero-price payoffs.

Using the pricing representation, define

$$r^* = \frac{p^*}{\pi(p^*)}. \tag{10}$$

Interpretation: The return r^* is the minimum conditional second-moment return. Every return can be decomposed into r^* plus a zero-cost payoff.

4.2 Conditional mean-variance frontier

The paper constructs $z^* \in Z$ satisfying

$$(z \mid z^*)_G = E(z \mid G), \quad z \in Z. \tag{11}$$

Then returns can be decomposed as

$$R = \{r : r = r^* + wz^* + n, \quad w \in I,\ n \in N\}, \tag{12}$$

where

$$N = \{z \in Z : E(z \mid G) = 0\}. \tag{13}$$

For a target conditional mean w, the frontier return is

$$r_w = r^* + w^* z^*, \quad w^* = \frac{w - E(r^* \mid G)}{E(z^* \mid G)}. \tag{14}$$

Interpretation: The conditional frontier is spanned by two conditional funds: r^* and z^*. Conditioning information changes the portfolio weights because w^* can be random.

4.3 Unconditional mean-variance frontier

Restrict to finite unconditional second-moment payoffs:

$$P^* = \{p \in P : E(p^2) < \infty\}. \tag{15}$$

Define

$$R^* = R \cap P^*, \tag{16}$$
$$Z^* = Z \cap P^*, \tag{17}$$
$$N^* = \{z \in Z^* : Ez = 0\}. \tag{18}$$

For a target unconditional mean c,

$$r_c = r^* + c^* z^*, \quad c^* = \frac{c - E(r^*)}{E(z^*)}. \tag{19}$$

Interpretation: The unconditional frontier uses constant weights, not state-contingent weights. Therefore the unconditional frontier is generally smaller than the conditional frontier.

4.4 Risk-free return and beta representation

If a unit payoff exists, the conditionally risk-free return is

$$r^f = \frac{1}{(p^* \mid 1)_G}. \tag{20}$$

The single-beta representation conditioned on G is

$$E(r \mid G) - \alpha = \frac{Cov(r_p, r \mid G)}{Var(r_p \mid G)}[E(r_p \mid G) - \alpha]. \tag{21}$$

Interpretation: A return can be conditionally mean-variance efficient without satisfying the unconditional beta relation used in standard regression tests.

5 Pseudo-pricing and unconditional moment restrictions

5.1 Pseudo-pricing function

To obtain restrictions in unconditional moments, the paper defines

$$\pi^*(p) = E[\pi(p)]. \tag{22}$$

If p^* has finite unconditional second moment, then

$$\pi^*(p) = E[(p \mid p^*)_G] = (p \mid p^*). \tag{23}$$

Interpretation: This converts the conditional pricing function into a real-valued pseudo-pricing function that can be tested with unconditional sample moments.

5.2 Moment-condition interpretation

The key empirical restriction is

$$E[\pi(p)] = E(pp^*) \tag{24}$$

for all admissible payoffs p.

Interpretation: This is the logic behind instrumental-variable or GMM tests of dynamic asset pricing models. Instruments form conditional portfolios; the econometrician tests unconditional moment restrictions.

5.3 Information loss

The paper shows that conditioning down alone does not necessarily lose information. If two conditional pricing functions imply the same pseudo-pricing function on a sufficiently rich payoff space, their benchmark payoffs are equal with probability one.

Interpretation: The main loss of power comes from omitting payoff spaces or instruments, not from unconditionalizing the pricing restrictions per se.

6 Identification and empirical design

6.1 What can be tested

- Asset pricing models imply pricing functions.
- Pricing functions imply restrictions on population moments of payoffs and prices.
- Under stationarity and ergodicity, those population moments can be estimated using time-series averages.
- Conditional restrictions can be converted into unconditional restrictions by multiplying payoffs by instruments and taking expectations.

Interpretation: This is the bridge from abstract dynamic theory to empirical asset pricing tests.

6.2 What goes wrong when conditioning information is omitted

- The unconditional frontier is not the same as the conditional frontier.
- A conditional reference return need not be an unconditional reference return.
- Standard regression tests can test unconditional efficiency but cannot automatically test conditional efficiency.
- More instruments may be needed to recover implications of the conditional model.

Interpretation: The paper warns against interpreting unconditional mean-variance tests as tests of conditional asset-pricing models unless the required conditions are verified.

6.3 Limitations

- The paper is theoretical and does not conduct a direct empirical test.
- The payoff space may be infinite-dimensional, so empirical tests necessarily use only a subset of assets and instruments.
- The theory assumes enough regularity for conditional Hilbert-space representations.
- Choosing a numeraire matters for the finite second-moment condition used in pseudo-pricing.

Interpretation: The paper does not eliminate the difficulty of empirical implementation; it clarifies exactly what restrictions empirical work is testing.

7 Tables, figures, and original equation panels

The original paper contains no empirical tables or plotted figures. To preserve the direct-extraction style, this section uses compact panels of original equations and theorem statements cropped directly from the paper.

7.1 Panel 1: Data generation and information sets

Read:

- The paper constructs stationary time series using a measure-preserving transformation S.
- Information at date t is generated by shifting the initial sigma algebra G.
- Payoffs and prices evolve through the same transformation, making initial-date pricing analysis applicable across time.

Economic message: The data-generation setup justifies using unconditional time-series averages to estimate population moments, even though agents condition on richer information.

time zero. Define

$$G_t = \{A_t : A_t = S^{-t}(A) \text{ for some } A \text{ in } G\}, \qquad \text{for} \qquad t = 1, 2, \ldots. \tag{1.2}$$

Then G_t is the sigma algebra defining the information available to economic agents at date t. We assume that G_1 contains G so that $\{G_t : t = 1, 2, \ldots\}$ is nondecreasing. Finally, for each $t = 1, 2, \ldots$, let I_t denote the set of all random variables that are measurable with respect to G_t and let I denote the set of all random variables that are measurable with respect to G.

At any date t economic agents make portfolio decisions based on their current information. Asset prices are assumed to be measurable with respect to G_t and hence observed by all consumers. A one-period security purchased at time t has a payoff at time $t+1$. The payoff for this security is assumed to be in the information set of economic agents at the date of payoff $(t+1)$. Let p denote a random variable that is in I_1 and is used to define a sequence of such payoffs. The corresponding payoff at time $t+1$ is given by

$$p_{t+1}(\omega) = p[S^t(\omega)]. \tag{1.3}$$

7.2 Panel 2: Conditional Hilbert-space primitives

Read:

- P^+ contains payoffs with finite conditional second moments.
- The conditional inner product is $(p_1 \mid p_2)_G = E(p_1 p_2 \mid G)$.
- The conditional norm is random and measurable with respect to G.

Economic message: This mathematical structure is what makes the pricing function a conditional analogue of a linear functional.

To describe these properties, it is convenient to introduce a conditional counterpart to an inner product on P^+. For any p_1 and p_2 in P^+, let

$$(2.2) \qquad \langle p_1 | p_2 \rangle_G = E(p_1 p_2 | G).$$

A conditional version of the Cauchy–Schwarz Inequality guarantees that the right-hand side of (2.2) is well-defined and finite.[3] In accordance with (2.2), we define a conditional norm to be

$$(2.3) \qquad \| p \|_G = [\langle p | p \rangle_G]^{1/2}.$$

Both the conditional inner product $\langle \cdot | \cdot \rangle_G$ and the conditional norm $\| \cdot \|_G$ map into I. To define convergence using these constructs, we must use a notion of convergence of random variables that are measurable with respect to G. We use convergence in probability in the following definitions of conditional Cauchy sequences and conditional convergent sequences.

Definition 2.1: A sequence $\{p_j : j = 1, 2, \ldots\}$ in P^+ converges conditionally to p_0 if for any $\varepsilon > 0$, $\lim_{j\to\infty} \Pr\{\| p_j - p_0 \|_G > \varepsilon\} = 0$.

Definition 2.2: A sequence $\{p_j : j = 1, 2, \ldots\}$ in P^+ is conditionally Cauchy if for any $\varepsilon > 0$, $\lim_{j,k\to\infty} \Pr\{\| p_j - p_k \|_G > \varepsilon\} = 0$.

Next, we introduce a subset P of P^+ that is restricted to satisfy conditional counterparts to linearity and completeness.

Definition 2.3: A set P is a conditional linear subspace of P^+ if for any w_1 and w_2 that are in I and any p_1 and p_2 in P, $w_1 p_1 + w_2 p_2$ is in P.

7.3 Panel 3: Conditional pricing representation

Read:

- Theorem 2.1 states that a continuous conditionally linear pricing function has a unique benchmark payoff p^*.
- All prices satisfy $\pi(p) = (p \mid p^*)_G$.
- Under no-arbitrage, p^* is positive.

Economic message: This is the paper's central pricing representation: dynamic asset pricing models can be indexed by their conditional stochastic discount factor.

will depend on the particular asset pricing model under investigation.

For our analysis, it is convenient to rule out certain nontrivial pricing functions.

ASSUMPTION 2.4: *There exists a payoff* p_0 *in* P *for which* $\Pr\{\pi(p_0)=0\}=0$.

A sufficient condition for Assumption 2.4 to be valid is that the pricing function π be strictly positive on P^+:

LEMMA 2.2: *Suppose* $P=P^+$, *Assumption* 2.2 *is satisfied, and* $\Pr\{\pi(p)>0\}=1$ *for any* p *in* P^+ *for which* $\Pr\{p>0\}=1$. *Then Assumptions* 2.3 *and* 2.4 *are satisfied.*

An example of a pricing function that satisfies Assumptions 2.2 and 2.3 is $\pi(p)=\langle p \mid p^*\rangle_G$ for some p^* in P. It turns out that all pricing functions have such a representation. This result is an implication of the conditional analogue to the Riesz Representation Theorem from the theory of Hilbert spaces.

THEOREM 2.1: *Suppose Assumptions* 2.1–2.4 *are satisfied. Then there exists a unique payoff* p^* *in* P *that satisfies* $\pi(p)=\langle p \mid p^*\rangle_G$ *for all* p *in* P. *Furthermore,* $\Pr\{\|p^*\|_G>0\}=1$.[5]

Theorem 2.1 shows that each asset pricing function is uniquely indexed by a

7.4 Panel 4: Conditional mean-variance decomposition

Read:

- The zero-price payoff space Z is decomposed into a conditional mean direction z^* and a conditional-zero-mean subspace N.
- Returns are represented as $r^* + wz^* + n$.
- Conditional frontier returns set $n = 0$.

Economic message: Conditioning information changes the efficient frontier because the weight w can depend on current information.

Theorem 2.1 also shows that $\langle z^* | z^* \rangle_G = E(z^{*2} | G) = E(z^* | G)$ is positive with probability one. Notice that

$$\text{(3.9)} \qquad \operatorname{Var}(z^* | G) = E(z^{*2} | G) - E(z^* | G)^2 = E(z^* | G)(1 - E(z^* | G)) \geq 0$$

implies that with probability one

$$\text{(3.10)} \qquad E(z^* | G) = \langle z^* | z^* \rangle_G \leq 1.$$

The random variable z^* defines one (conditional) dimension of Z. All remaining dimensions that are orthogonal to z^* conditioned on G must have conditional mean zero since (3.8) is satisfied. Hence, the set Z can be represented as

$$\text{(3.11)} \qquad Z = \{z\colon z = wz^* + n \text{ for some } w \text{ in } I \text{ and some } n \text{ in } N\}$$

where N is the set

$$\text{(3.12)} \qquad N = \{z \text{ in } Z\colon E(z | G) = 0\}.$$

Combining (3.7) and (3.11) gives the representation

$$\text{(3.13)} \qquad R = \{r\colon r = r^* + wz^* + n \text{ for some } w \text{ in } I \text{ and some } n \text{ in } N\}.$$

Next, we consider solutions to the mean-variance problem conditioned on G.

PROBLEM 3.1: Minimize $\langle r | r \rangle_G$ for r in R subject to the constraint $E(r | G) = w$ for some w in I.

Notice that the objective function for Problem 3.1 is not real-valued but is a random variable in I. Hence, this objective only induces a partial ordering on P. Nevertheless, as we show in the subsequent analysis, Problem 3.1 does have a solution. Also, notice that $\langle r | r \rangle_G$ in Problem 3.1 could be replaced by the variance of r conditioned on G without altering the solution. Any return that solves Problem 3.1 for some w in I is said to be on the mean-variance frontier conditioned on G.

Problem 3.1 can be solved conveniently using the decomposition of returns given in (3.13). Let

$$\text{(3.14)} \qquad w^* = [w - E(r^* | G)]/E(z^* | G).$$

Then r in R satisfies $E(r | G) = w$ if, and only if,

$$\text{(3.15)} \qquad r = r^* + w^* z^* + n$$

for w^* given by (3.14) and some n in N. To solve Problem 3.1, n in (3.15) is set to zero since the components on the right-hand side of (3.15) are orthogonal conditioned on G. We have proved the following lemma.

LEMMA 3.3: *If (P, π) satisfies Assumptions 2.1–2.4 and $[Z, E(\cdot | G)]$ satisfies Assumption 3.1, then $r_w = r^* + w^* z^*$ is the solution to Problem 3.1 for w^* given by (3.14).*

7.5 Panel 5: Unconditional frontier and risk-free return

Read:

- The unconditional frontier uses R^*, Z^*, and N^*, the finite-unconditional-second-moment counterparts.
- The frontier return is $r_c = r^* + c^* z^*$ with constant c^*.
- A risk-free return lies on the unconditional frontier only under special conditions, such as a constant risk-free rate.

Economic message: The unconditional frontier is not equivalent to the conditional frontier. Omitting information imposes an additional restriction: frontier weights must be constant rather than state-contingent.

Any solution to Problem 3.2 for some real number c is said to be on the unconditional mean-variance frontier. Our strategy for calculating this frontier is very similar to the one used to calculate the mean-variance frontier conditioned on G. Applying the Law of Iterated Expectations to (3.8) gives

(3.21) $\quad \langle z|z^*\rangle = E(z) \quad$ for all z in Z^*.

Hence, the random variable z^* is orthogonal to N^* (unconditionally) so that elements in Z^* can be represented as the sum of a scalar multiple of z^* and an element in N^*. Consequently, R^* can be represented as

(3.22) $\quad R^* = \{r\colon r = r^* + c^*z^* + n^*$ for some real number c^* and some n^* in $N^*\}$.

Let

(3.23) $\quad c^* = [c - E(r^*)]/E(z^*)$.

Then r in R^* has expectation c if, and only if,

(3.24) $\quad r = r^* + c^*z^* + n^*$

for some n^* in N^* and c^* given in (3.23). Problem 3.2 is then solved by setting n^* to zero. We have proved the following lemma.

LEMMA 3.4: *Suppose that* (P, π) *satisfies Assumptions* 2.1–2.4 *and* 3.2, *and* $[Z, E(\cdot|G)]$ *satisfies Assumption* 3.1. *Then the solution to Problem* 3.2 *is* $r_c = r^* + c^*z^*$ *for* c^* *given by* (3.23).

Taken together, Lemmas 3.3 and 3.4 display the impact of omitting conditioning information when evaluating whether a return is on the mean-variance frontier. Notice that any return that is on the unconditional frontier must also be on the conditional frontier. In other words, r_c given in Lemma 3.4 also solves Problem 3.1 for w equal to $E(r_c|G)$. The converse is not true, however. A return may be on the mean-variance frontier conditioned on G and not be on the unconditional frontier. For instance, consider r_w as given by Lemma 3.3. Then as long as w^* as given by (3.14) is not equal to a constant with probability one, r_w will not be on the unconditional mean-variance frontier. A risk-free return can be used to illustrate this fact.

When P contains a unit payoff and π has no arbitrage opportunities on P, R will contain a return that is risk-free. This return is given by

(3.25)
$$r^f = 1/\langle p^*|1\rangle_G = \langle r^*|r^*\rangle_G/\langle r^*|1\rangle_G.$$

Notice that the conditional inner product of any payoff in P with a unit payoff is just the conditional expectation of that payoff. In this case, the variable z^* used in representing the mean-variance frontier is the conditional residual of regressing 1 on r^*. Hence

(3.26) $\quad z^* = 1 - \langle r^*|1\rangle_G r^*/\langle r^*|r^*\rangle_G$.

7.6 Panel 6: Conditional and unconditional single-beta representations

Read:

- A conditional reference return satisfies a conditional beta representation.
- The unconditional analogue requires constant weights on the unconditional frontier.
- Corollary 3.1 characterizes when a return supports an unconditional single-beta representation.

Economic message: Regression-based beta tests are valid for unconditional mean-variance efficiency, but they do not automatically test conditional efficiency.

It is straightforward to verify that z^* satisfies (3.8). The risk-free return can be expressed as

$$r^f = r^* + r^f z^*. \tag{3.27}$$

This means that the risk-free return will be on the unconditional mean-variance frontier if, and only if, r^f equals a constant with probability one.

The implications of the static CAPM are typically portrayed using a single-beta representation of the risk-return tradeoff. We call a return r_β in R a *reference return* for a single-beta representation conditioned on G if $\Pr\{\operatorname{Var}(r_\beta \mid G) = 0\} = 0$ and

$$E(r \mid G) - \alpha = \frac{\operatorname{Cov}[r_\beta, r \mid G]}{\operatorname{Var}[r_\beta \mid G]}[E(r_\beta \mid G) - \alpha] \tag{3.28}$$

for all r in R, where the random variable α is in I. In general, any return in R that is conditionally uncorrelated with r_β has a conditional mean equal to α. In particular, if a risk-free return r^f is in I, then $r^f = \alpha$ since $\operatorname{Cov}[r_\beta, r^f \mid G] = 0$.

Our next result establishes the conditions under which being on the conditional mean-variance frontier is equivalent to being a reference return for a conditional single-beta representation. It is the conditional counterpart of Roll's (1977) Corollary 6.

LEMMA 3.5: *Suppose (P, π) satisfies Assumption 2.1–2.4, π has no arbitrage opportunities on P, and $[Z, E(\cdot \mid G)]$ satisfies Assumption 3.1. Then r_β is a reference return for a single-beta representation conditioned on G if, and only if, $r_\beta = r^* + w^* z^*$ where w^* is in I and*

$$\Pr\{w^* = E(r^* \mid G)/(1 - E(z^* \mid G))\} = 0. \tag{3.29}$$

The no-arbitrage restriction on π ensures that $(1 - E(z^* \mid G))$ is not zero with probability one. Condition (3.29) guarantees that the probability that r_β is equal to the minimum conditional variance return is zero.

We call a return r_β a reference return for an unconditional single-beta representation if $\operatorname{Var}(r_\beta) > 0$ and the unconditional counterpart to (3.28) is satisfied for all returns in R^*, where α is a real number. Notice that the unconditional version of (3.28) defines a set of restrictions across the means of returns and the population regression coefficients of returns on r_β. There is also an unconditional version of Lemma 3.5.

COROLLARY 3.1: *Suppose the assumptions of Lemma 3.5 and Assumption 3.2 are satisfied. Then r_β is a reference return for an unconditional single-beta representation if, and only if, $r_\beta = r^* + c^* z^*$ where c^* is a constant and*

$$c^* \neq E(r^*)/(1 - E(z^*)). \tag{3.30}$$

7.7 Panel 7: Pseudo-pricing and moment restrictions

Read:

- The pseudo-pricing function is $\pi^*(p) = E[\pi(p)]$.
- Theorem 4.1 shows that the pseudo-pricing function satisfies the standard pricing-function assumptions with the trivial sigma algebra.
- The same p^* represents both conditional prices and pseudo-prices.

Economic message: This is the paper's route from conditional asset-pricing theory to unconditional moment restrictions used in empirical work.

to be valid, we require that the payoff p^*, used in representing π, be in P^*.

ASSUMPTION 4.1: $\langle p^* | p^* \rangle < \infty$.

Assumption 4.1 will not be satisfied for all choices of numeraire for quoting asset prices. More precisely, let w be a random variable in I for which $\Pr\{w>0\} = 1$. Then a pricing function π^- with a different numeraire is given by $\pi^-(p) = \pi(p)/w$. This pricing function has a benchmark payoff p^*/w. If p^* satisfies Assumption 4.1, then it is not necessarily the case that p^*/w will satisfy Assumption 4.1. On the other hand, if p^* does not satisfy Assumption 4.1 then one can always find a random variable w such that p^*/w will satisfy Assumption 4.1. For instance, let $w = \|p^*\|_G$. Then $E[(p^*/w)^2 | G] = 1$ implying that p^*/w has a finite unconditional second moment. Therefore, requiring a pricing function to satisfy this assumption restricts the choice of numeraire.

We define π^* to be

$$\pi^*(p) = E[\pi(p)]. \tag{4.1}$$

Notice that π^* maps into the real numbers. This function behaves like the pricing function π when the trivial sigma algebra is used for the conditioning information set. The only random variables that are measurable with respect to the trivial sigma algebra are constant over all sample points. We refer to prices calculated using π^* as pseudo-prices.

THEOREM 4.1: *Suppose* (P, π) *satisfies Assumptions* 2.1–2.4 *and* 4.1. *Then* (P^*, π^*) *satisfies Assumptions* 2.1–2.4 *for G given by the trivial sigma algebra. Furthermore, if* π *has no arbitrage opportunities on P, then* π^* *has no arbitrage opportunities on* P^*.

Figure 1: Pseudo-pricing function and theorem.

CONDITIONING INFORMATION 605

so that the inner product representation

$$\pi^*(p) = E[\langle p | p^* \rangle_G] = \langle p | p^* \rangle \tag{4.2}$$

is valid. Hence, p^* can be used to represent π^* as well as π. This feature is attractive for the following reason. Suppose time series data are available on p^* as implied by a particular asset pricing model. In addition, suppose time series data are available on asset payoffs and prices. Under the data generation processes described in Section 1, both $E[\pi(p)]$ and $E(pp^*)$ can be estimated for any p in P^*. Relations (4.1) and (4.2) imply that these two quantities should be the same. Consequently, the asset pricing model can be tested by checking whether sample counterparts to $E[\pi(p)]$ and $E(pp^*)$ could be equal after accounting for estimation error. This procedure can be viewed as an interpretation of the econometric approach suggested by Hansen and Singleton (1982) for testing intertemporal asset pricing models.

Notice that payoffs can be constructed that are conditional linear combinations of some initial collection of payoffs as long as the coefficients are measurable with respect to G. The prices (using π as the pricing function) of the resulting payoffs are just the conditional linear combinations of the corresponding prices (again using π) of the initial collection of payoffs. Hence, an analyst is given great flexibility in constructing payoffs and prices to be used in this procedure. The conditional weights used in forming payoffs and prices correspond to the instrumental variables in the Hansen–Singleton analysis. Hansen and Singleton show formally how to estimate preference parameters and test restrictions using time series versions of instrumental variables methods.

A question emerges as to whether information is lost in testing the implications using π^* instead of π. One way to think about this problem is to ask when can two pricing functions, say π and π^+, defined on P imply the same pseudo pricing function on P^*. Suppose these two pricing functions have the representations

$$\pi(p) = \langle p | p^* \rangle_G \quad \text{for all } p \text{ in } P \tag{4.3}$$

and

$$\pi^+(p) = \langle p | p^+ \rangle_G \quad \text{for all } p \text{ in } P, \tag{4.4}$$

where both p^* and p^+ are in P. In addition, suppose p^* and p^+ are in P^* and π and π^+ imply the same pseudo pricing function:

$$\pi^*(p) = E[\pi(p)] = E[\pi^+(p)] \tag{4.5}$$

for all p in P^*. Then

Figure 2: Uniqueness and information-loss argument.

8 Short summary

- The paper studies how conditioning information affects testable implications of dynamic asset pricing models.
- Prices and payoffs are both random variables; prices are constrained by current information and payoffs by future information.
- Under conditional linearity and continuity, the pricing function has a conditional inner-product representation.
- The unique benchmark payoff p^* is the conditional stochastic discount factor.
- Mean-variance implications are sensitive to conditioning information.
- The unconditional frontier is generally smaller than the conditional frontier.
- A conditionally efficient return need not be unconditionally efficient.
- Standard beta regressions test unconditional frontier restrictions, not conditional frontier restrictions.
- The pseudo-pricing function $\pi^*(p) = E[\pi(p)]$ gives unconditional moment restrictions that can be estimated from time series.
- The Hansen–Singleton empirical strategy can be interpreted through this pseudo-pricing representation.
- Omitting payoff spaces or instruments can reduce the power of asset pricing tests.

9 Conclusion

9.1 Core conclusion

Hansen and Richard show that conditioning information is not a technical detail; it changes the interpretation of testable restrictions in asset pricing. Dynamic models imply conditional pricing functions, and these functions are best represented using conditional Hilbert-space methods.

9.2 Economic intuition

Economic agents price assets using information available at the trading date. An econometrician often observes time-series averages and tests unconditional moment restrictions. The paper explains how to translate conditional pricing restrictions into unconditional restrictions without confusing conditional and unconditional efficiency concepts.

9.3 Takeaway

The key lesson is that omitting conditioning information can change the object being tested. Unconditional beta tests and mean-variance tests are valid for unconditional restrictions, but they are not automatic tests of the conditional restrictions implied by dynamic asset pricing models. Proper empirical tests require either modeling conditioning information or using instruments to encode it in unconditional moments.